Optimizing International Trade Tariff Subsidies via Finite Mathematics and Cooperative Game Theory

Abstract: *This report details the deployment of finite mathematics, specifically covariance matrix algebras, continued fraction expansions, and Shapley value allocations, to optimize international trade dynamics. Focusing on the substitution of traditional agreements (e.g., CETA) across grain, beef, and mineral ore sectors, we demonstrate mathematically verifiable, fair-trade structures using non-zero-sum cooperative game computations.*

1. Introduction

Modern international trade introduces complex multi-agent tariff negotiations. By modeling trade as a cooperative game, we can ensure surplus value is distributed symmetrically proportional to sectoral equity. This guarantees a mathematically fair model for Canada and the EU.

2. Finite Mathematics in Portfolio Allocations

The allocation of trade dividends across Grain, Beef, and Mineral Ore relies on finite state arrays and covariance matrices. Expected portfolio models act over finite states to minimize variance. By computing the portfolio volatility as the dot product of transposed weight vectors and finite covariance matrices, decision-makers map deterministic substitution risks directly into optimal dividend allocations.

3. Cooperative Trade Subventions via Shapley Value

Subsidies and tariffs form a coalition game. The Shapley value provides a unique mapping derived from finite factorial summations over all possible coalition permutations. This calculation dictates equitable extraction of tariffs between Canada and the EU, negating arbitrarily weighted trade wars and isolating each participant's marginal market contribution.

4. Verifiability using Continued Fractions

Trade exchange ratios (e.g., Beef volume relative to Mineral Ore volume) are scrutinized using algorithms generating continued fraction expansions. The principal convergents of this expansion algorithmically limit bounds of rational tariff approximations, cementing a mathematically verifiable transaction ledger impervious to floating-point manipulation or asymmetric inflation.

5. Econophysics and Optimal Resource Allocation

Expanding the finite analytical model, we introduce econophysics to conceptualize trade networks as aggregate thermodynamic interactions. Employing the Boltzmann-Gibbs entropy maximization principle:

$$S = -k_B \Sigma p_i \ln(p_i)$$

we derive optimal global resource distributions. In these scenarios, trade sectors act as collision particles exchanging economic 'energy' (capital/resources), maintaining a socio-economic equilibrium defined by exponential wealth distributions. This allows robust predictions of resource liquidity across Canada and EU networks.

6. Conclusion

Deploying computational finite mathematics establishes absolute transparency and Pareto optimality within bilateral trade ecosystems. This algorithmic construct provides an immediate, empirically verifiable baseline for modern non-coercive global trade implementations.